

- 1 A student makes measurements from which he calculates the speed of sound in air as  $335.61 \text{ m s}^{-1}$ . He estimates the percentage uncertainty in the speed of sound to be  $\pm 4.5\%$ .

How should the student record the speed of sound, considering the uncertainty?

- A  $330 \text{ m s}^{-1}$       B  $335.6 \text{ m s}^{-1}$       C  $336 \text{ m s}^{-1}$       D  $340 \text{ m s}^{-1}$

(D)

- 2 The Stefan-Boltzmann law for the rate of thermal energy emitted per unit surface area of a body is given by

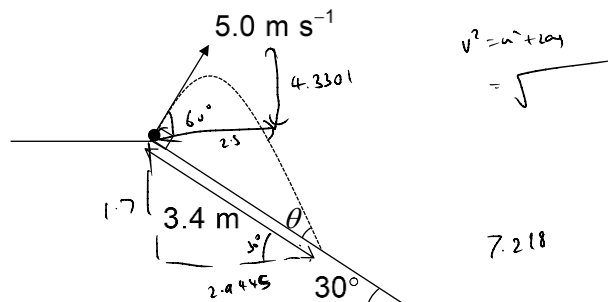
$$\frac{P}{A} = e\sigma T^4,$$

where  $P$  is the rate of thermal energy emitted,  $A$  is the surface area,  $T$  is the thermodynamic temperature and  $e$ , a dimensionless constant, is the emissivity of the body and  $\sigma$  is the Stefan-Boltzmann constant.

Which is the S.I. base unit for  $\sigma$ ?

- A  $\text{kg s}^{-3} \text{ K}^{-4}$       B  $\text{kg s}^{-1} \text{ K}^4$       C  $\text{W m}^{-2} \text{ K}^{-4}$       D  $\text{W m}^{-2} \text{ }^\circ\text{C}^{-4}$

- 3 A stone is projected perpendicularly to a slope with a velocity of  $5.0 \text{ m s}^{-1}$  and lands  $3.4 \text{ m}$  down the slope. The slope is inclined at an angle of  $30^\circ$  to the horizontal. Air resistance is negligible.



What is the angle  $\theta$  that the velocity of the stone makes with the slope as it lands?

- A  $27^\circ$       B  $32^\circ$       C  $41^\circ$       D  $71^\circ$

$\tan^{-1} \left( \frac{7.21}{2.445} \right)$